

No Uniform Interpolation for Linear Logic

Rémi Di Guardia

IRIF – Inria, Université Paris Cité, France

Lyon, 24 June 2025



Interpolation

Definition: Craig's Interpolation

An **interpolant** for two formulas A and B with $A \vdash B$ is a formula C s.t.:

- $\text{Voc}(C) \subseteq \text{Voc}(A) \cap \text{Voc}(B)$
- $A \vdash C$ is provable
- $C \vdash B$ is provable

$$\forall A, \forall B, \exists C, \dots$$

Interpolation

Definition: Craig's Interpolation

An **interpolant** for two formulas A and B with $A \vdash B$ is a formula C s.t.:

- $\text{Voc}(C) \subseteq \text{Voc}(A) \cap \text{Voc}(B)$
- $A \vdash C$ is provable
- $C \vdash B$ is provable

$$\forall A, \forall B, \exists C, \dots$$

$$\forall A, \exists C, \forall B, \dots$$

Definition: Uniform Interpolation

A **uniform interpolant** for a formula A and an atom X is a formula C s.t.:

- $\text{Voc}(C) \subseteq \text{Voc}(A) \setminus \{X\}$
- $A \vdash C$ is provable
- for all formulas B such that $X \notin \text{Voc}(B)$ and $A \vdash B$ is provable, $C \vdash B$ is also provable

Interpolation

Definition: Craig's Interpolation

An **interpolant** for two formulas A and B with $A \vdash B$ is a formula C s.t.:

- $\text{Voc}(C) \subseteq \text{Voc}(A) \cap \text{Voc}(B)$
- $A \vdash C$ is provable
- $C \vdash B$ is provable

$$\forall A, \forall B, \exists C, \dots$$

$$\forall A, \exists C, \forall B, \dots$$

Definition: Uniform Interpolation

A **uniform interpolant** for a formula A and an atom X is a formula C s.t.:

- $\text{Voc}(C) \subseteq \text{Voc}(A) \setminus \{X\}$
- $A \vdash C$ is provable
- for all formulas B such that $X \notin \text{Voc}(B)$ and $A \vdash B$ is provable,
 $C \vdash B$ is also provable

$\approx \exists$ right introduction

$\approx \exists$ left introduction

Some Logics with(out) Uniform Interpolation

Classical Logic



Easy

A

$$\leadsto A[T/X] \vee A[\perp/X]$$

Some Logics with(out) Uniform Interpolation

Classical Logic



Easy

$$A \quad \rightsquigarrow \quad A[\top/X] \vee A[\perp/X]$$

Intuitionistic Logic



[Pitts 1992]

Syntactic approach, contraction rule problematic

$$(X \rightarrow Y) \rightarrow Z \quad \rightsquigarrow \quad ((Y \rightarrow Z) \rightarrow Y) \rightarrow Z$$

Some Logics with(out) Uniform Interpolation

Classical Logic



Easy

$$A \quad \rightsquigarrow \quad A[\top/X] \vee A[\perp/X]$$

Intuitionistic Logic



[Pitts 1992]

Syntactic approach, contraction rule problematic

$$(X \rightarrow Y) \rightarrow Z \quad \rightsquigarrow \quad ((Y \rightarrow Z) \rightarrow Y) \rightarrow Z$$

Modal Logic S4



[Ghilardi and Zawadowski 1995]

Explicit **counter-example** with a game model

Some Logics with(out) Uniform Interpolation

Classical Logic



Easy

$$A \quad \rightsquigarrow \quad A[\top/X] \vee A[\perp/X]$$

Intuitionistic Logic



[Pitts 1992]

Syntactic approach, contraction rule problematic

$$(X \rightarrow Y) \rightarrow Z \quad \rightsquigarrow \quad ((Y \rightarrow Z) \rightarrow Y) \rightarrow Z$$

Modal Logic S4



[Ghilardi and Zawadowski 1995]

Explicit **counter-example** with a game model

(I)MALL



[Alizadeh, Derakhshan, and Ono 2014]

Same **syntactic** method as for Intuitionistic Logic

$$X^+ \otimes (X^+ \multimap Y^+) \quad \rightsquigarrow \quad Y^+$$

Some Logics with(out) Uniform Interpolation

Classical Logic



Easy

$$A \quad \rightsquigarrow \quad A[\top/X] \vee A[\perp/X]$$

Intuitionistic Logic



[Pitts 1992]

Syntactic approach, contraction rule problematic

$$(X \rightarrow Y) \rightarrow Z \quad \rightsquigarrow \quad ((Y \rightarrow Z) \rightarrow Y) \rightarrow Z$$

Modal Logic S4



[Ghilardi and Zawadowski 1995]

Explicit **counter-example** with a game model

(I)MALL



[Alizadeh, Derakhshan, and Ono 2014]

Same **syntactic** method as for Intuitionistic Logic

$$X^+ \otimes (X^+ \multimap Y^+) \quad \rightsquigarrow \quad Y^+$$

(I)LL



[this talk]

Explicit **counter-example** with a formula

Linear Logic (1/2)

Formulas

$A, B ::=$	$ X^- X^+$	<i>atom</i>
	$ A \wp B A \otimes B \perp 1$	<i>multiplicative</i>
	$ A \oplus B A \& B 0 \top$	<i>additive</i>
	$?A !A$	<i>exponential</i>

Involutive Negation

$$\begin{aligned}(X^-)^\perp &= X^+ & (X^+)^\perp &= X^- \\ (A \wp B)^\perp &= A^\perp \otimes B^\perp & (A \otimes B)^\perp &= A^\perp \wp B^\perp & \perp^\perp &= 1 & 1^\perp &= \perp \\ (A \oplus B)^\perp &= A^\perp \& B^\perp & (A \& B)^\perp &= A^\perp \oplus B^\perp & 0^\perp &= \top & \top^\perp &= 0 \\ (?A)^\perp &= !A^\perp & (!A)^\perp &= ?A^\perp\end{aligned}$$

Linear Arrow

$$A \multimap B = A^\perp \wp B$$

Linear Logic (2/2)

Rules

$$\begin{array}{c}
\frac{}{\vdash X^-, X^+} \text{ (ax)} \qquad \frac{\vdash A^\perp, \Gamma \quad \vdash A, \Delta}{\vdash \Gamma, \Delta} \text{ (cut)} \\
\\
\frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} \text{ (}\wp\text{)} \qquad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \text{ (}\otimes\text{)} \qquad \frac{\vdash \Gamma}{\vdash \perp, \Gamma} \text{ (}\perp\text{)} \qquad \frac{}{\vdash 1} \text{ (1)} \\
\\
\frac{\vdash A, \Gamma \quad \vdash B, \Gamma}{\vdash A \& B, \Gamma} \text{ (}\&\text{)} \qquad \frac{\vdash A, \Gamma}{\vdash A \oplus B, \Gamma} \text{ (}\oplus_1\text{)} \qquad \frac{\vdash B, \Gamma}{\vdash A \oplus B, \Gamma} \text{ (}\oplus_2\text{)} \qquad \frac{}{\vdash \top, \Gamma} \text{ (}\top\text{)} \\
\\
\frac{\vdash A, \Gamma}{\vdash ?A, \Gamma} \text{ (?d)} \qquad \frac{\vdash ?A, ?A, \Gamma}{\vdash ?A, \Gamma} \text{ (?c)} \qquad \frac{\vdash \Gamma}{\vdash ?A, \Gamma} \text{ (?w)} \qquad \frac{\vdash A, ?\Gamma}{\vdash !A, ?\Gamma} \text{ (!)}
\end{array}$$

Cut-Elimination

Definition: Uniform Interpolation

Uniform Interpolant for a formula A and an atom X : formula C s.t.

- $\text{Voc}(C) \subseteq \text{Voc}(A) \setminus \{X\}$
- $\vdash A^\perp, C$ is provable
- for all formulas B such that $X \notin \text{Voc}(B)$ and $\vdash A^\perp, B$ is provable, $\vdash C^\perp, B$ is also provable

The Counter-Example

Goal: Exhibit formulas A and $(B_n)_{n \in \mathbb{N}}$ s.t.:

- 1 $X \in \text{Voc}(A)$ and $X \notin \text{Voc}(B_n)$
- 2 $\vdash A^\perp, B_n$ is provable
- 3 for all formulas C such that $X \notin \text{Voc}(C)$,
 $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ both provable $\implies \text{size}(C) \geq n$

The Counter-Example

Goal: Exhibit formulas A and $(B_n)_{n \in \mathbb{N}}$ s.t.:

- 1 $X \in \text{Voc}(A)$ and $X \notin \text{Voc}(B_n)$
- 2 $\vdash A^\perp, B_n$ is provable
- 3 for all formulas C such that $X \notin \text{Voc}(C)$,
 $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ both provable $\implies \text{size}(C) \geq n$

Solution:

$$A = X^+ \otimes ! (X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$$
$$B_n = \underbrace{Y^+ \otimes \dots \otimes Y^+}_{n+1 \text{ times}}$$

The Counter-Example

Goal: Exhibit formulas A and $(B_n)_{n \in \mathbb{N}}$ s.t.:

- 1 $X \in \text{Voc}(A)$ and $X \notin \text{Voc}(B_n)$
- 2 $\vdash A^\perp$, B_n is provable
- 3 for all formulas C such that $X \notin \text{Voc}(C)$,
 $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ both provable $\implies \text{size}(C) \geq n$

Solution:

$$A = X^+ \otimes ! (X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$$
$$B_n = \underbrace{Y^+ \otimes \dots \otimes Y^+}_{n+1 \text{ times}}$$

Idea:

- 2 Feed successively X^+ to n copies of $! (X^+ \multimap (X^+ \otimes Y^+))$ to get n times Y^+ , then consume X^+ through $(X^+ \multimap Y^+)$.

The Counter-Example

Goal: Exhibit formulas A and $(B_n)_{n \in \mathbb{N}}$ s.t.:

- 1 $X \in \text{Voc}(A)$ and $X \notin \text{Voc}(B_n)$
- 2 $\vdash A^\perp$, B_n is provable
- 3 for all formulas C such that $X \notin \text{Voc}(C)$,
 $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ both provable $\implies \text{size}(C) \geq n$

Solution:

$$A = X^+ \otimes ! (X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$$
$$B_n = \underbrace{Y^+ \otimes \dots \otimes Y^+}_{n+1 \text{ times}}$$

Idea:

- 2 Feed successively X^+ to n copies of $!(X^+ \multimap (X^+ \otimes Y^+))$ to get n times Y^+ , then consume X^+ through $(X^+ \multimap Y^+)$.
- 3 “ C has no useful ! on Y^+ ” since $!(X^+ \multimap (X^+ \otimes Y^+))$ from A cannot really be used above a promotion. Hence “ $C^\perp$ has no useful ? on Y^- ” so it has $(n+1)$ different Y^- to link to the $(n+1)$ Y^+ of B_n .

Case in point: candidate $!Y^+ \otimes Y^+$ but $\vdash A^\perp, !Y^+ \otimes Y^+$ **not** provable!

The easy part: 1 and 2

Lemma 1

$$X \in \text{Voc}(A) \text{ and } X \notin \text{Voc}(B_n) \text{ for all } n \in \mathbb{N}$$

Proof.

Trivial.



Lemma 2

$\vdash A^\perp, B_n$ is provable (intuitionistically) for all $n \in \mathbb{N}$

Proof.

Provide the proof trees by an easy induction on n .

$$\begin{array}{c}
\frac{}{\vdash X^-, X^+} \text{ (ax)} \quad \frac{}{\vdash Y^-, Y^+} \text{ (ax)} \\
\frac{}{\vdash X^-, X^+} \text{ (ax)} \quad \frac{}{\vdash Y^-, Y^-, Y^+ \otimes Y^+} \text{ (}\otimes\text{)} \\
\frac{}{\vdash X^-, X^+} \text{ (ax)} \quad \frac{}{\vdash X^-, Y^-, X^+ \otimes Y^-, Y^+ \otimes Y^+} \text{ (}\otimes\text{)} \\
\frac{}{\vdash X^-, X^+} \text{ (ax)} \quad \frac{}{\vdash X^- \wp Y^-, X^+ \otimes Y^-, Y^+ \otimes Y^+} \text{ (}\wp\text{)} \\
\frac{}{\vdash X^-, X^+ \otimes (X^- \wp Y^-), X^+ \otimes Y^-, Y^+ \otimes Y^+} \text{ (}\otimes\text{)} \\
\frac{}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)), X^+ \otimes Y^-, Y^+ \otimes Y^+} \text{ (?}_d\text{)} \\
\frac{}{\vdash X^-, ?(X^+ \otimes (X^- \wp Y^-)) \wp (X^+ \otimes Y^-), Y^+ \otimes Y^+} \text{ (}\wp\text{)} \\
\frac{}{\vdash X^- \wp ?(X^+ \otimes (X^- \wp Y^-)) \wp (X^+ \otimes Y^-), Y^+ \otimes Y^+} \text{ (}\wp\text{)}
\end{array}$$



The hard part: 3

Idea: “ C has no useful $!$ on Y^+ ” for $!(X^+ \multimap (X^+ \otimes Y^+))$ from A is useless without some X^+ . Hence, $C^\perp$ contains a Y^- for each Y^+ in B_n .

Lemma: “ C has no useful $!$ on Y^+ ”

Let C be such that $X \notin \text{Voc}(C)$ and $\vdash A^\perp, C$ is provable. There is a cut-free proof of $\vdash A^\perp, C$ where **!-rules have no X in their conclusions**.

Proof.

Not complicated: do some substitutions $X \mapsto 0$, some compositions by *cut*, and use the sub-formula property. □

The hard part: 3

Idea: “ C has no useful $!$ on Y^+ ” for $!(X^+ \multimap (X^+ \otimes Y^+))$ from A is useless without some X^+ . Hence, $C^\perp$ contains a Y^- for each Y^+ in B_n .

Lemma: “ C has no useful $!$ on Y^+ ”

Let C be such that $X \notin \text{Voc}(C)$ and $\vdash A^\perp, C$ is provable. There is a cut-free proof of $\vdash A^\perp, C$ where **$!$ -rules have no X in their conclusions.**

Proof.

Not complicated: do some substitutions $X \mapsto 0$, some compositions by *cut*, and use the sub-formula property. $\square$

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof?

Yep. Indeed.

How to **propagate** the data from the previous result to *ax*-rules on B_n ?

Geometry Of Interaction in poor man's proof-nets

Definition: GOI projection graph $\mathcal{G}_\pi$ of a proof π of $\vdash \Gamma$

Undirected edge-bicoloured graph defined as:

Example

π	$\mathcal{G}_\pi$
$ \frac{}{\vdash X^-, X^+} (ax) \quad \frac{}{\vdash ?X^-, X^+} (?_d) \quad \frac{}{\vdash X^-, X^+} (ax) \quad \frac{}{\vdash ?X^-, X^+} (?_d) $	$ \frac{}{\vdash X^-, X^+} (ax) \quad \frac{}{\vdash X^-, X^+} (ax) $
$ \frac{}{\vdash ?X^-, X^+} (?_d) \quad \frac{}{\vdash X^-, X^+} (ax) \quad \frac{}{\vdash ?X^-, X^+} (?_d) \quad \frac{}{\vdash ?X^-, X^+} (?_d) $	$ \frac{}{\vdash ?X^-, X^+} (?_d) \quad \frac{}{\vdash ?X^-, X^+} (?_d) $
$ \frac{}{\vdash ?X^-, !X^+} (!) \quad \frac{}{\vdash ?X^-, X^+} (?_d) \quad \frac{}{\vdash ?X^-, ?X^-, X^+ \otimes X^+} (\otimes) $	$ \frac{}{\vdash ?X^-, X^+ \otimes X^+} (?_c) $
$ \frac{}{\vdash ?X^-, !X^+ \& X^+} (\&) \quad \frac{}{\vdash ?X^-, X^+ \otimes X^+} (\otimes_1) $	$ \frac{}{\vdash ?X^- \oplus X^-, X^+ \otimes X^+} (\oplus_1) $
$ \frac{}{\vdash ?X^-, X^+ \otimes X^+} (cut) $	

Geometry Of Interaction in poor man's proof-nets

Definition: GOI projection graph $\mathcal{G}_\pi$ of a proof π of $\vdash \Gamma$

Undirected edge-bicoloured graph defined as:

Vertices: Occurrences of atomic formulas from Γ and the *cut*-formulas

Example

π	$\mathcal{G}_\pi$
$\frac{\frac{\frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, X^+}^{(?_d)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, !X^+}^{(!)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, X^+ \otimes X^+}^{(?_d)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, X^+ \otimes X^+}^{(?_d)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, !X^+ \& X^+}^{(\&)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, X^+ \otimes X^+}^{(cut)}}{\vdash ?X^-, X^+ \otimes X^+}^{(\&)} \quad \frac{\overline{\vdash X^-, X^+}^{(ax)}}{\vdash ?X^-, X^+ \otimes X^+}^{(cut)}$	$\underbrace{X^-}_{?X^-} \quad \underbrace{X^+ \quad X^+}_{cut\text{-formula} \quad !X^+ \& X^+} \quad \underbrace{X^- \quad X^-}_{cut\text{-formula} \quad ?X^- \oplus X^-} \quad \underbrace{X^+ \quad X^+}_{X^+ \otimes X^+}$

Geometry Of Interaction in poor man's proof-nets

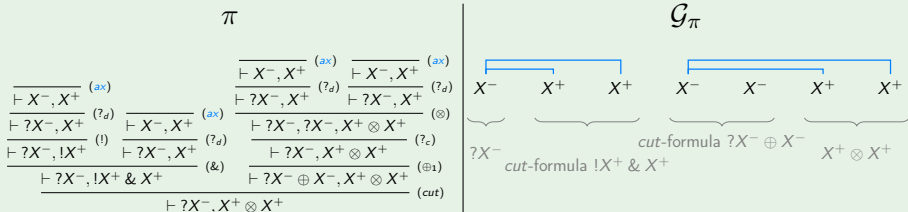
Definition: GOI projection graph $\mathcal{G}_\pi$ of a proof π of $\vdash \Gamma$

Undirected edge-bicoloured graph defined as:

Vertices: Occurrences of atomic formulas from Γ and the *cut*-formulas

ax-edges: Between occurrences with an *ax*-rule on them in π

Example



Geometry Of Interaction in poor man's proof-nets

Definition: GOI projection graph $\mathcal{G}_\pi$ of a proof π of $\vdash \Gamma$

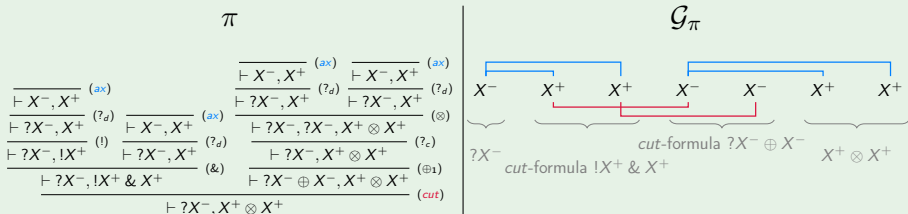
Undirected edge-bicoloured graph defined as:

Vertices: Occurrences of atomic formulas from Γ and the *cut*-formulas

ax-edges: Between occurrences with an *ax*-rule on them in π

cut-edges: Between dual occurrences of a *cut*-formula

Example



Geometry Of Interaction in poor man's proof-nets

Definition: GOI projection graph $\mathcal{G}_\pi$ of a proof π of $\vdash \Gamma$

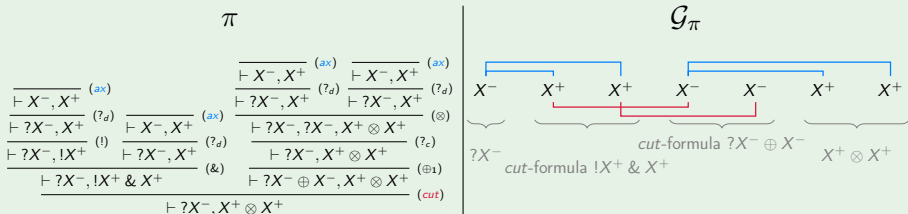
Undirected edge-bicoloured graph defined as:

Vertices: Occurrences of atomic formulas from Γ and the *cut*-formulas

ax-edges: Between occurrences with an *ax*-rule on them in π

cut-edges: Between dual occurrences of a *cut*-formula

Example



Several **ax-edges** on an occurrence: either due to a $?_c$ -rule or to a $\&$ -rule

Alternating persistent paths

Alternating path: consecutive edges are coloured differently

Lemma

Assume $\pi \longrightarrow \rho$. If there is an alternating path between two vertices in $\mathcal{G}_\rho$, then there also is an alternating path between them in $\mathcal{G}_\pi$.

Proof.

Simple inspection on the kind of the cut-elimination step. □

Alternating persistent paths

Alternating path: consecutive edges are coloured differently

Lemma

Assume $\pi \longrightarrow \rho$. If there is an alternating path between two vertices in $\mathcal{G}_\rho$, then there also is an alternating path between them in $\mathcal{G}_\pi$.

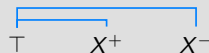
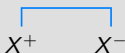
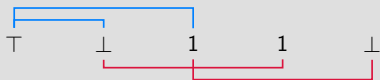
Proof.

Simple inspection on the kind of the cut-elimination step. □

$\top$ Warning $\perp$

Adding units and **T-edges** for $\top$ -rules is a **bad** idea: the above lemma fails!

$$\frac{\frac{\frac{}{\vdash \top, \perp \wp 1} (\top)}{\vdash \top, X^+, X^-} \quad \frac{\frac{\frac{}{\vdash 1} (1) \quad \frac{\frac{}{\vdash X^+, X^-} (ax)}{\vdash \perp, X^+, X^-} (\perp)}{\vdash 1 \otimes \perp, X^+, X^-} (\otimes)}{\vdash \top, X^+, X^-} (cut)
 }{\vdash \top, X^+, X^-} (\top)$$



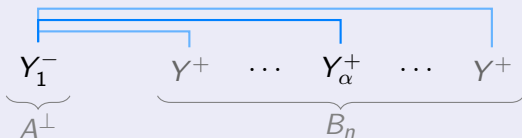
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-



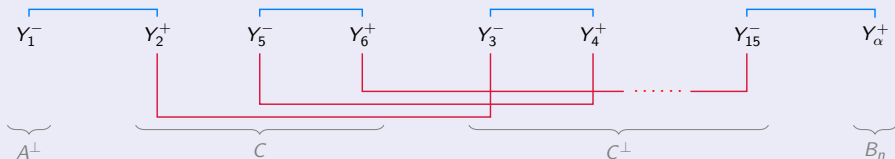
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*



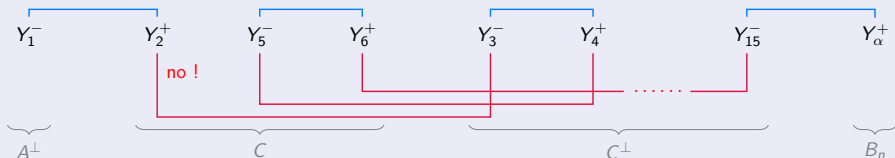
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



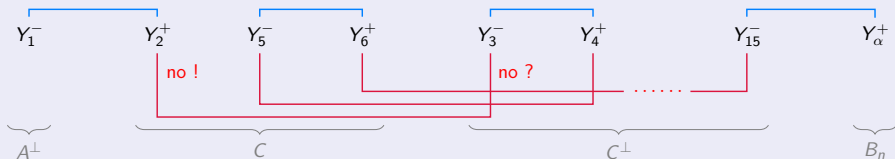
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



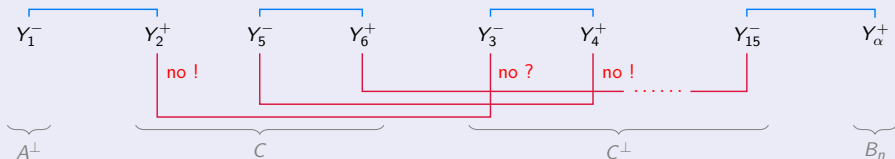
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



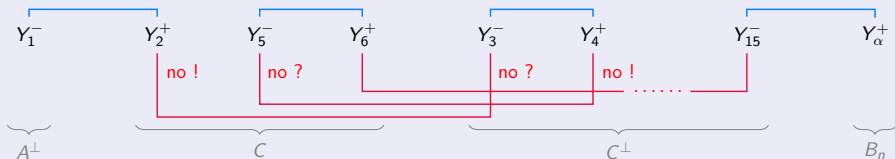
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



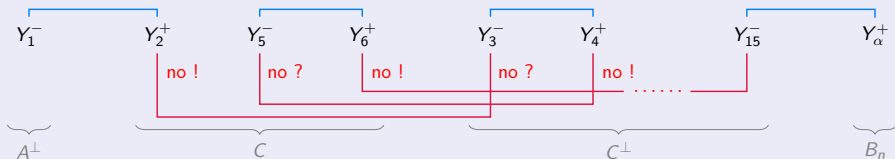
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



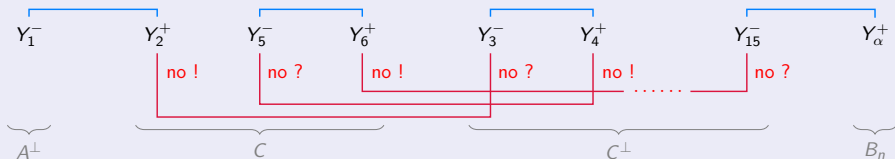
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**



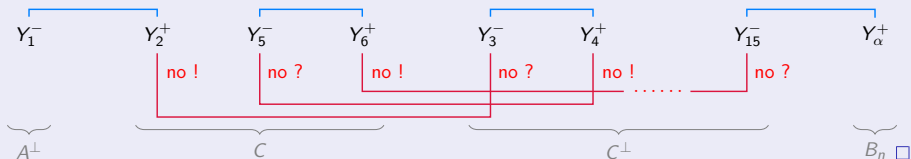
Back to Lemma 3

Lemma 3

Let C be a formula with $X \notin \text{Voc}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are provable, then C contains (at least) $n + 1$ occurrences of Y^+ .

Proof.

- 1 Cut + normalise the two proofs: ax -rules with each Y^+ and a Y^-
- 2 These are **alternating paths**: we track each back *before normalising*
- 3 Lemma “ C has no useful ! on Y ”; **propagation along the path**
- 4 Do everything from the start in a **0-slice**: take the proof and delete one sub-tree of every $\&$ -rule that is not above a $!$ -rule



& Warning !

We can adapt everything done before with 0-slices but not with slices.

Conclusion

Proposition

Linear Logic does not have uniform interpolation.

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$ □

Conclusion

Proposition

(Intuitionistic) Linear Logic does not have uniform interpolation.

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$ □

Conclusion

Proposition

(Intuitionistic) Linear / Affine Logic does not have uniform interpolation.

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$
as no weakening on sub-formulas of B_n in all affine proofs of $\vdash A^\perp, B_n$ $\square$

Conclusion

Proposition

(Intuitionistic) Linear / Affine Logic does not have uniform interpolation.

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$
as no weakening on sub-formulas of B_n in all affine proofs of $\vdash A^\perp, B_n$ $\square$

Observations

- Why a counter-example for **Linear Logic** and not for *Classical Logic*, or *Intuitionistic Logic*, or *MALL*?
→ A with a sub-formula that can be contracted, and one that cannot

Conclusion

Proposition

(*Intuitionistic*) *Linear / Affine Logic does not have uniform interpolation.*

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$
as no weakening on sub-formulas of B_n in all affine proofs of $\vdash A^\perp, B_n$ $\square$

Observations

- Why a counter-example for **Linear Logic** and not for *Classical Logic*, or *Intuitionistic Logic*, or *MALL*?
→ A with a sub-formula that can be contracted, and one that cannot
- Problem about **provability** but *GOI* and *cut-elimination* very useful
→ Proof of Lemma 3 without GOI projection graphs?

Conclusion

Proposition

(*Intuitionistic*) *Linear / Affine Logic does not have uniform interpolation.*

Proof.

No interpolant for $A = X^+ \otimes !(X^+ \multimap (X^+ \otimes Y^+)) \otimes (X^+ \multimap Y^+)$
as no weakening on sub-formulas of B_n in all affine proofs of $\vdash A^\perp, B_n$ $\square$

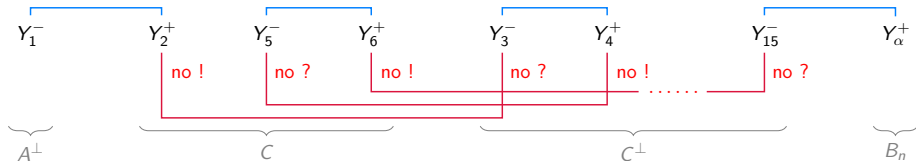
Observations

- Why a counter-example for **Linear Logic** and not for *Classical Logic*, or *Intuitionistic Logic*, or *MALL*?
→ A with a sub-formula that can be contracted, and one that cannot
- Problem about **provability** but *GOI* and *cut-elimination* very useful
→ Proof of Lemma 3 without GOI projection graphs?

Perspectives

- **Decision problem**: given some A , does it have a uniform interpolant?
- Proof simple enough to be **formalized** in Rocq?

Thank you for
your attention!



References



Alizadeh, Majid, Farzaneh Derakhshan, and Hiroakira Ono (2014). “Uniform interpolation in substructural logics”. In: *The Review of Symbolic Logic* 7.3, pp. 455–483. DOI: [10.1017/S175502031400015X](https://doi.org/10.1017/S175502031400015X).



Ghilardi, Silvio and Marek Zawadowski (June 1995). “Undefinability of propositional quantifiers in the modal system S4”. In: *Studia Logica* 55.2, pp. 259–271. DOI: [10.1007/BF01061237](https://doi.org/10.1007/BF01061237).



Pitts, Andrew M. (1992). “On an interpretation of second order quantification in first order intuitionistic propositional logic”. In: *Journal of Symbolic Logic* 57.1, pp. 33–52. DOI: [10.2307/2275175](https://doi.org/10.2307/2275175).

Back-up: Proof of “ C has no useful ! on Y^+ ”

Lemma: “ C has no useful ! on Y^+ ”

Let C be such that $X \notin \text{Voc}(C)$ and $\vdash A^\perp, C$ is provable. There is a cut-free proof of $\vdash A^\perp, C$ where !-rules have no X in their conclusions.

Proof.

Let π be a cut-free derivation of $\vdash A^\perp, C$ with a minimal number of !-rules having $?(X^+ \otimes (X^- \wp X^-))$ in their conclusion sequents. We prove there are none. By the sub-formula property, any given !-rule is of the shape

$$\frac{\rho \quad \vdash (?(X^+ \otimes (X^- \wp Y^-)))^i, D, ?\Delta}{\vdash (?(X^+ \otimes (X^- \wp Y^-)))^i, !D, ?\Delta} (!)$$

where $!D, ?\Delta$ are sub-formulas of C . If $i \geq 1$, then we contradict the minimality of π with π where we replace this sub-tree with a normal form of

$$\frac{\frac{\frac{\vdash \top, 0 \otimes Y^+}{\vdash \top \wp (0 \otimes Y^+)} (\top) \quad \frac{\vdash \top \wp (0 \otimes Y^+)}{\vdash !(\top \wp (0 \otimes Y^+))} (!)}{\vdash !(\top \wp (0 \otimes Y^+))} (\top) \quad \frac{\rho[0/X] \quad \vdash (?(0 \otimes (\top \wp Y^-)))^i, D, ?\Delta}{\vdash ?(0 \otimes (\top \wp Y^-)), D, ?\Delta} (\wp_c)}{\vdash D, ?\Delta} (cut) \quad \frac{\vdash !D, ?\Delta}{\vdash !D, ?\Delta} (!) \quad \frac{\vdash !D, ?\Delta}{\vdash (?(X^+ \otimes (X^- \wp Y^-)))^i, !D, ?\Delta} (?_w)$$

□

Back-up: 0-slice = slice at depth 0

Definition: 0-slice

A 0-slice of a proof π is the result of deleting one of the two sub-trees of each $\&$ -rule of π that is not above a !-rule. Some $\&$ -rules become unary:

$$\frac{\vdash A, \Gamma}{\vdash A \& B, \Gamma} (\&_1) \qquad \frac{\vdash B, \Gamma}{\vdash A \& B, \Gamma} (\&_2)$$

Example

The diagram illustrates the construction of a 0-slice from a proof tree. A jagged line indicates the deletion of sub-trees not above !-rules.

Original Proof Tree (Left):

- Top level: $\frac{\vdash \top, !(1\&1)}{\vdash \top \& \perp, !(1\&1)} (\top)$
- Second level (under $\vdash \top \& \perp, !(1\&1)$):
 - Left branch: $\frac{\frac{\frac{\overline{\vdash 1}^{(1)}}{\vdash 1\&1} (\&)}{\vdash !(1\&1)} (!)}{\vdash \perp, !(1\&1)} (\perp)$
 - Right branch: $\frac{\overline{\vdash 1}^{(1)}}{\vdash \perp, !(1\&1)} (\&)$

0-Slice (Right):

- Top level: $\frac{\overline{\vdash \top, !(1\&1)}}{\vdash \top \& \perp, !(1\&1)} (\&_1)$
- Second level (under $\vdash \top \& \perp, !(1\&1)$):
 - Left branch: $\frac{\frac{\overline{\vdash 1}^{(1)}}{\vdash 1\&1} (\&)}{\vdash !(1\&1)} (!)$
 - Right branch: $\frac{\overline{\vdash 1}^{(1)}}{\vdash \perp, !(1\&1)} (\perp)$
- Bottom level: $\frac{\vdash \top \& \perp, !(1\&1)}{\vdash \top \& \perp, !(1\&1)} (\&_2)$

Back-up: 0-slice and cut-elimination

Cut-elimination extends to 0-slices in the natural way:

$$\frac{\frac{\frac{\vdash \Gamma, A}{\vdash \Gamma, A \& B} \text{ } (\&_1) \quad \frac{\frac{\vdash A^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} \text{ } (\oplus_1)}{\vdash \Gamma, \Delta} \text{ } (cut) \quad \longrightarrow \quad \frac{\vdash \Gamma, A \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} \text{ } (cut)$$
$$\frac{\frac{\frac{\vdash \Gamma, A}{\vdash \Gamma, A \& B} \text{ } (\&_1) \quad \frac{\frac{\vdash B^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta} \text{ } (\oplus_2)}{\vdash \Gamma, \Delta} \text{ } (cut) \quad \not\longrightarrow$$

Lemma

If $\pi \longrightarrow \rho$, then for each 0-slice ρ' of ρ , there exists a 0-slice π' of π such that $\pi' \longrightarrow \rho'$ or $\pi' = \rho'$.

Proof.

Simple inspection on the kind of the cut-elimination step. □

& Warning !

The above result holds for 0-slices but not for slices: failure for a $?_c - !$ key cut-elimination case. See for instance $\overline{?(A \oplus B)} \simeq ?A \wp ?B$.

Back-up: Affine Logic

Same as Linear Logic except:

$$\begin{array}{c}
 \frac{\frac{\frac{\vdash \Gamma}{\vdash ?A, \Gamma} (?w)}{\vdash \Gamma, ?\Delta} \quad \frac{\frac{\vdash A^\perp, ?\Delta}{\vdash !A^\perp, ?\Delta} (!)}{\vdash \Gamma, ?\Delta} (cut) \rightarrow \frac{\frac{\vdash \Gamma}{\vdash \Gamma, ?\Delta} (?w)}{\vdash \Gamma, \Delta} (w) \sim \frac{\frac{\vdash \Gamma}{\vdash A, \Gamma} (w)}{\vdash \Gamma, \Delta} (w) \vdash A^\perp, \Delta (cut) \rightarrow \frac{\frac{\vdash \Gamma}{\vdash \Gamma, \Delta} (w)}{\vdash \Gamma, \Delta} (w)
 \end{array}$$

$$\begin{array}{c}
 \frac{\frac{\frac{\vdash A, \Gamma}{\vdash A, ?B, \Gamma} (?w)}{\vdash \Gamma, ?B, \Delta} \quad \frac{\vdash A^\perp, \Delta}{\vdash \Gamma, ?B, \Delta} (cut) \rightarrow \frac{\frac{\vdash A, \Gamma}{\vdash \Gamma, \Delta} \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta} (cut) \sim \frac{\frac{\vdash A, \Gamma}{\vdash A, B, \Gamma} (w)}{\vdash \Gamma, B, \Delta} (w) \vdash A^\perp, \Delta (cut) \rightarrow \frac{\frac{\vdash A, \Gamma}{\vdash \Gamma, \Delta} \quad \vdash A^\perp, \Delta}{\vdash \Gamma, B, \Delta} (w) (cut)
 \end{array}$$

Lemma

Cut-elimination in Affine Logic is weakly normalizing.

Proof.

Simple adaptation of such a proof for Linear Logic. □

Back-up: No Uniform Interpolation in Affine Logic

Lemma

In Affine Logic there is no proof of

$X^a, (X \multimap (X \otimes Y))^b, (! (X \multimap (X \otimes Y)))^c, (X \otimes Y)^d, Y^e, (X \multimap Y)^f \vdash$

Proof.

By contradiction, consider a smallest cut-free proof. Its last rule is either a $\otimes$ - $\wp$ - $?d$ - $?c$ - or w -rule, with as premise a sequent of the same kind. $\square$

Proposition

(Intuitionistic) Affine Logic does not have uniform interpolation.

Proof.

Same proof as for Linear Logic, except we use the above lemma so as to have an ax -rule on each Y^+ of B_n , hence an alternating path in our GOI projection graph. $\square$